

Induction Motor Calculation Analysis

Mathematical Derivations & Parameter Mapping for ASYN4 Simulation Data

MOTOR DESIGNATION
SC315L 90KC10

RATED POWER (PN)
90.0 kW

RATED VOLTAGE (UN)
415 V

POLE PAIRS (P)
5 (10-Pole)

1. PRIMARY OPERATING SPECIFICATIONS

The primary ratings originate directly from input file EINASYN.DAT parameters and fundamental machine relationships.

Synchronous Speed (n_{syn})

$$n_{syn} = (60 \times f_N) / p = (60 \times 50.0) / 5 = 600.0 \text{ rpm}$$

Where f_N = 50.0 Hz (Supply Frequency) and p = 5 (Pole Pair Count).

Rated Speed (n_N)

$$n_N = n_{syn} \times (1 - S_{4/4}) = 600 \times (1 - 0.01075) = 593.55 \text{ rpm}$$

Derived using full-load fractional slip $S_{4/4}$ = 0.01075 (Output report: 593.5 rpm).

Rated Current (I_N)

$$I_N = P_N / (\sqrt{3} \times U_N \times \eta \times \cos\phi)$$

$$I_N = 90,000 / (\sqrt{3} \times 415 \times 0.9415 \times 0.789) \approx 168.0 \text{ A}$$

Evaluated at 4/4 nominal operating point with efficiency η = 94.15% and power factor $\cos\phi$ = 0.789.

2. LOAD PERFORMANCE PROFILE (P / P_N)

Performance metrics calculated across standard fractional load steps:

Load Step (P/P _N)	1/4 Load	2/4 Load	3/4 Load	4/4 Load	5/4 Load
Power Factor ($\cos\phi$)	0.414	0.633	0.740	0.789	0.808
Efficiency (η %)	88.30%	92.83%	93.97%	94.15%	93.87%
Rotor Slip (S)	0.00253	0.00509	0.00781	0.01075	0.01400
Calculated Speed (rpm)	598.5	596.9	595.3	593.55	591.6

3. STARTING & PEAK TORQUE CHARACTERISTICS

Starting Current Ratio (I_A / I_N)

$$I_A / I_N = 1016.5 \text{ A} / 168.0 \text{ A} = 6.05$$

Based on full voltage locked-rotor current (1016.5 A).

Starting Torque Ratio (M_A / M_N)

$$M_N = (9549 \times 90) / 593.5 = 1448 \text{ Nm}$$

$$M_A / M_N = 4158 \text{ Nm} / 1448 \text{ Nm} = 2.87$$

Direct ratio of locked-rotor torque to rated torque.

4. SHORT-CIRCUIT CHARACTERISTICS (KURZSCHLUSS-KENNLINIE)

Values evaluated from locked-rotor simulation tests across variable voltage steps. Standard electrical power relationships govern each line:

Apparent Power (S_A) & Active Power (P_A)

$$S_A = \sqrt{3} \times U_I \times I_A \quad (kVA)$$

$$P_A = S_A \times \cos\varphi_A \quad (kW)$$

Example at $U_1 = 415\text{ V}$: $S_A = \sqrt{3} \times 415 \times 1016.5 = 730.7\text{ kVA}$; $P_A = 730.7 \times 0.449 = 328.2\text{ kW}$.

U1 (V)	IA (A)	MA (Nm)	cos φA	PA (kW)	SA (kVA)	KKA
41.5	90.7	33	0.399	2.6	6.5	0.84
124.5	279.4	315	0.412	24.8	60.3	0.73
207.5	476.1	917	0.422	72.2	171.1	0.65
290.5	680.5	1876	0.432	147.8	342.4	0.58
373.5	897.6	3258	0.442	256.9	580.7	0.53
415.0	1016.5	4158	0.449	328.2	730.7	0.50

5. NO-LOAD DATA ANALYSIS (LEERLAUF - WARME MASCHINE)

The total no-load input power P_0 represents the accumulation of core losses, friction, windage, and stray residual losses:

No-Load Power Balance

$$P_0 = P_{FE} + P_{RBG} + P_{mech}$$

$$2.701\text{ kW} = 1.730\text{ kW (Iron)} + 0.600\text{ kW (Friction/Windage)} + 0.371\text{ kW (Mechanical/Stray)}$$

No-load current $I_0 = 76.30\text{ A}$ with no-load power factor $\cos\varphi_0 = 0.049$.

6. THERMAL & DYNAMIC PARAMETERS

Temperature Rises

$$\Delta T_1 = 55.2\text{ K (Stator)}$$

$$\Delta T_2 = 31.0\text{ K (Rotor)}$$

Measured at rated operational load.

No-Load Acceleration Time

$$t_{acc} = 0.2\text{ s}$$

$$J_{motor} = 9.91\text{ kg}\cdot\text{m}^2$$

Run-up time at nominal supply voltage.

Stator Phase Resistance

$$R_1 = 0.0493\text{ }\Omega \text{ (at } 20^\circ\text{C)}$$

Used as baseline parameter for equivalent circuit computations in ASYN4.